

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data – Renewable Energy | Data: Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

The urgency for a rapid shift to low-carbon energy sources

75%

of global GHG emissions
from fossil fuels

5M+

premature deaths per year
from fossil-fuel air pollution

>150 yrs

of fossil-fuel-dominated
energy since the Industrial Revolution

Since the Industrial Revolution, most countries' energy systems have become dominated by fossil fuels. Fossil fuel combustion is the primary driver of climate change and is also responsible for local air pollution that causes at least 5 million premature deaths per year.

To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies.

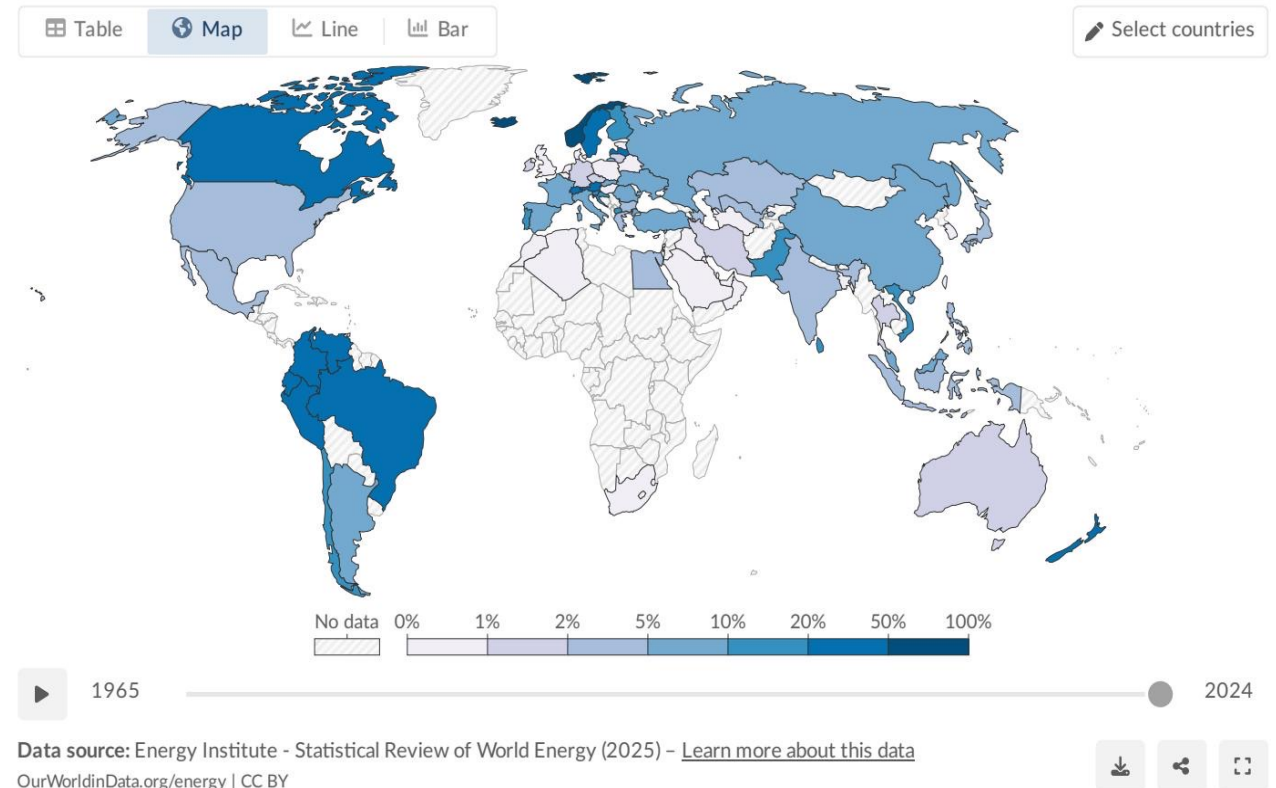
Renewables Now Supply ~14% of Global Primary Energy

2024 data, substitution method | Primary energy = electricity + transport + heating

~14% (one-seventh) of global primary energy comes from renewables

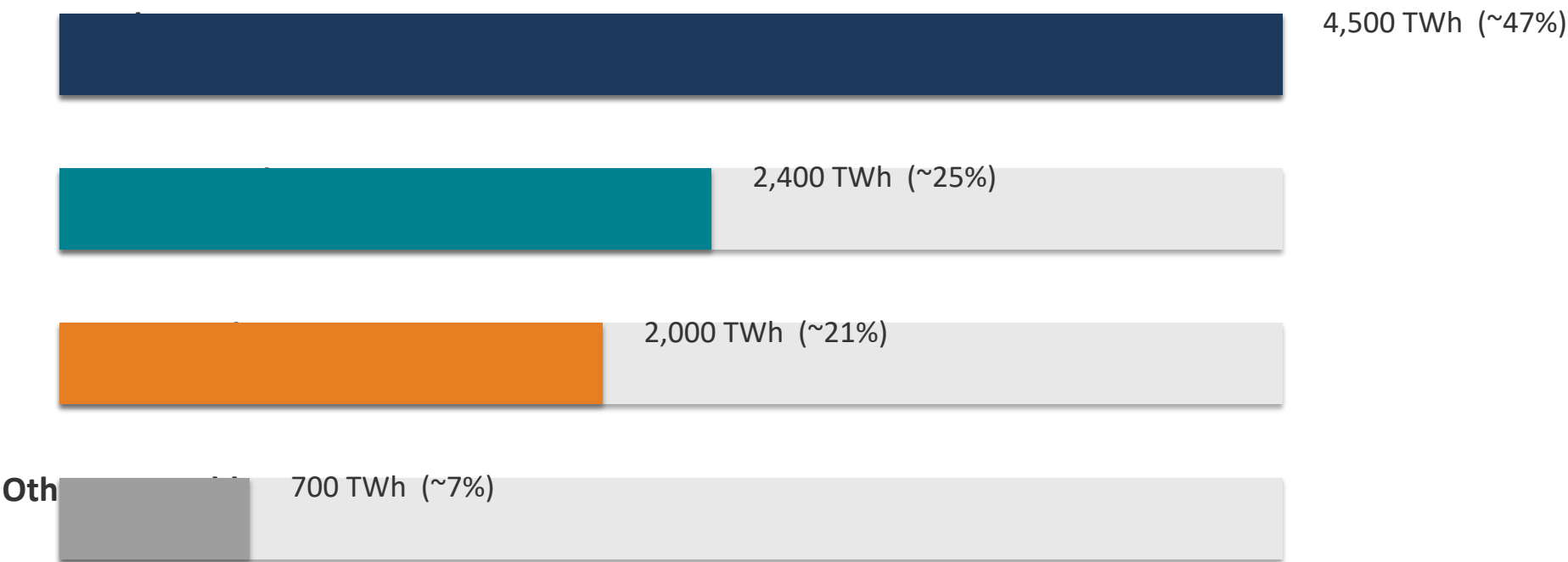
The renewable share in the total energy mix is lower than in the electricity mix alone because transport and heating are harder to decarbonize and remain more reliant on oil and gas.

Regional variation is significant: Norway, Brazil, and New Zealand derive large shares of primary energy from renewables, while fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5%.



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global renewable electricity generation by technology, 2024



Wind and solar combined (~4,400 TWh) now nearly match hydropower’s output (~4,500 TWh), and their growth rates far exceed hydro’s. The acceleration in wind and solar generation since ~2010 has been driven by rapidly falling costs and strong policy support.

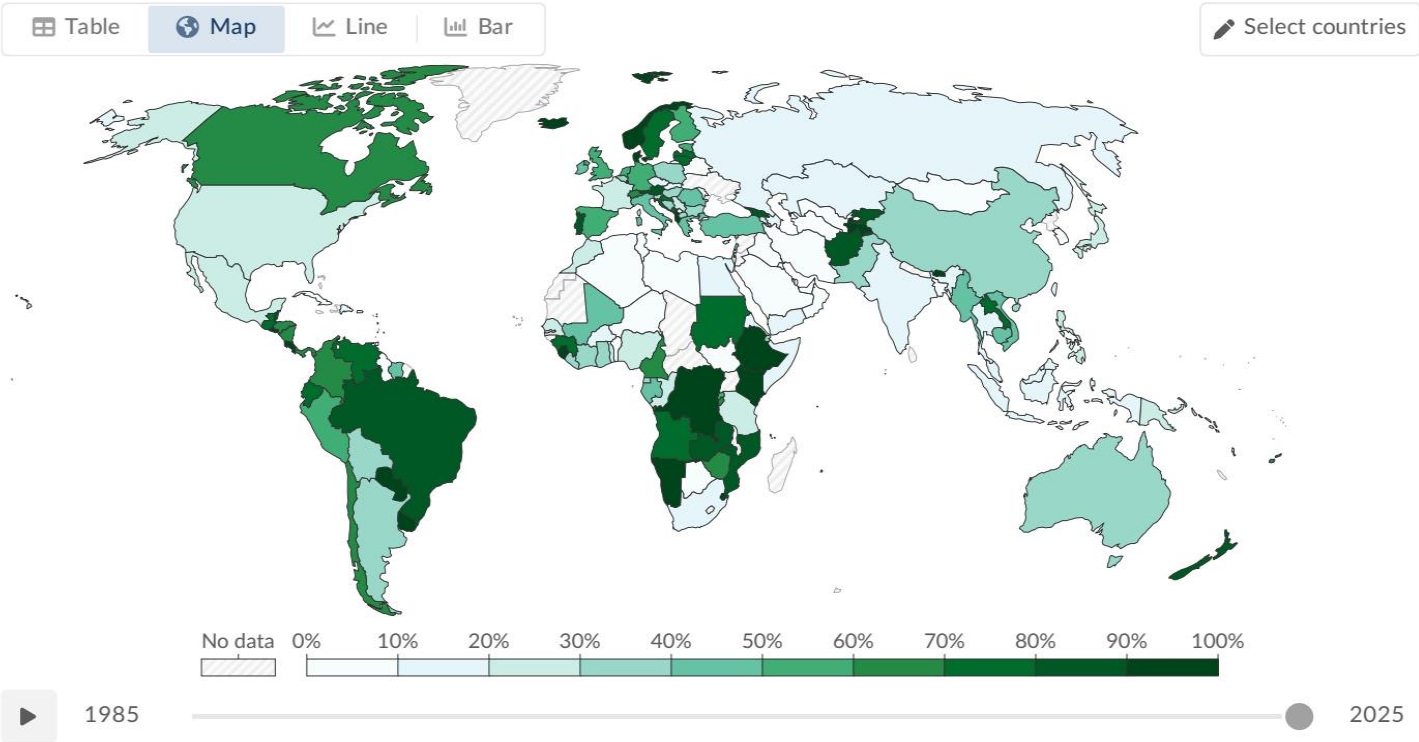
Other = geothermal, biomass, wave, tidal

Almost One-Third of Global Electricity Now Comes from Renewables

Renewable share of electricity by country, 2025

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Several African and Middle Eastern countries have renewable electricity shares below 10%, reflecting both fossil fuel dependence and limited investment in renewables.

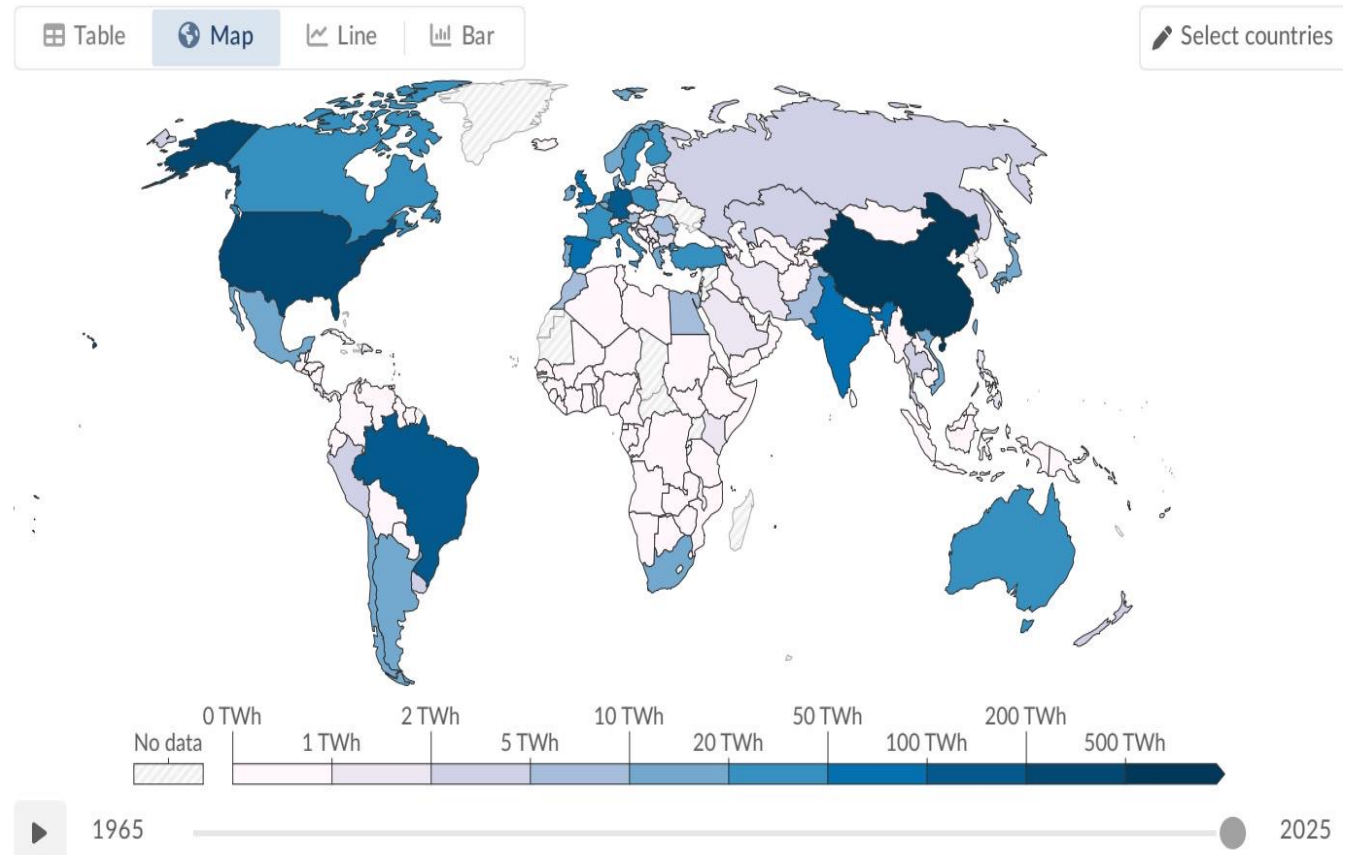
World average: ~30%

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower has generated electricity at scale for over a century

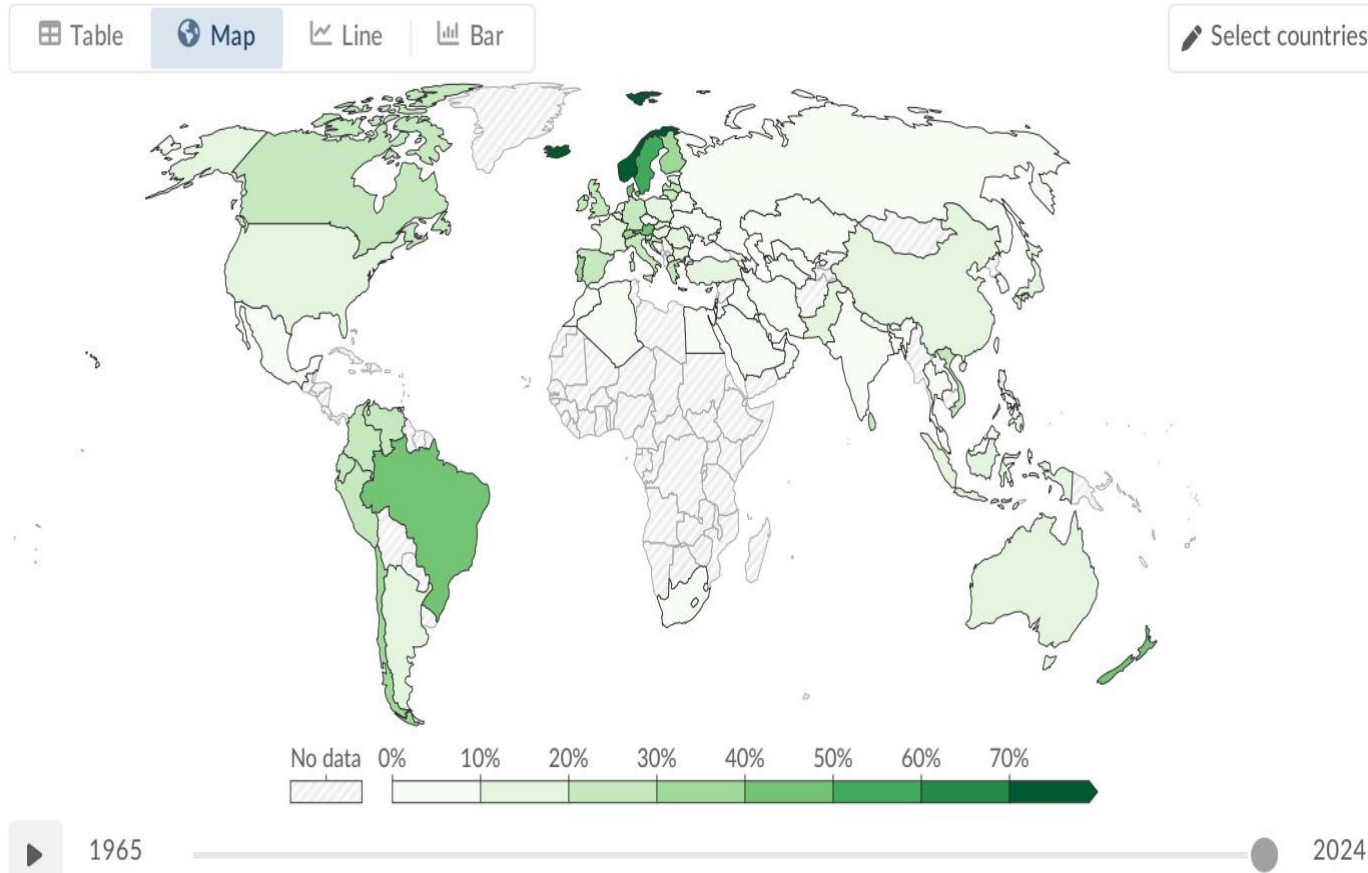
~4,500 TWh | ~47–50% of renewable electricity

- Remains the single largest renewable electricity source
- Has been generating electricity at scale for over a century
- Top producers: China, Brazil, Canada
- Growth potential is limited compared to wind and solar — most suitable sites already developed



Wind and Solar Are the Fastest-Growing Energy Technologies in History

Rapid growth since ~2010 driven by falling costs and policy support



WIND

~2,400 TWh generation

~25% of renewable electricity

Major contributor in Europe, China, and the U.S.

SOLAR

~2,000 TWh generation

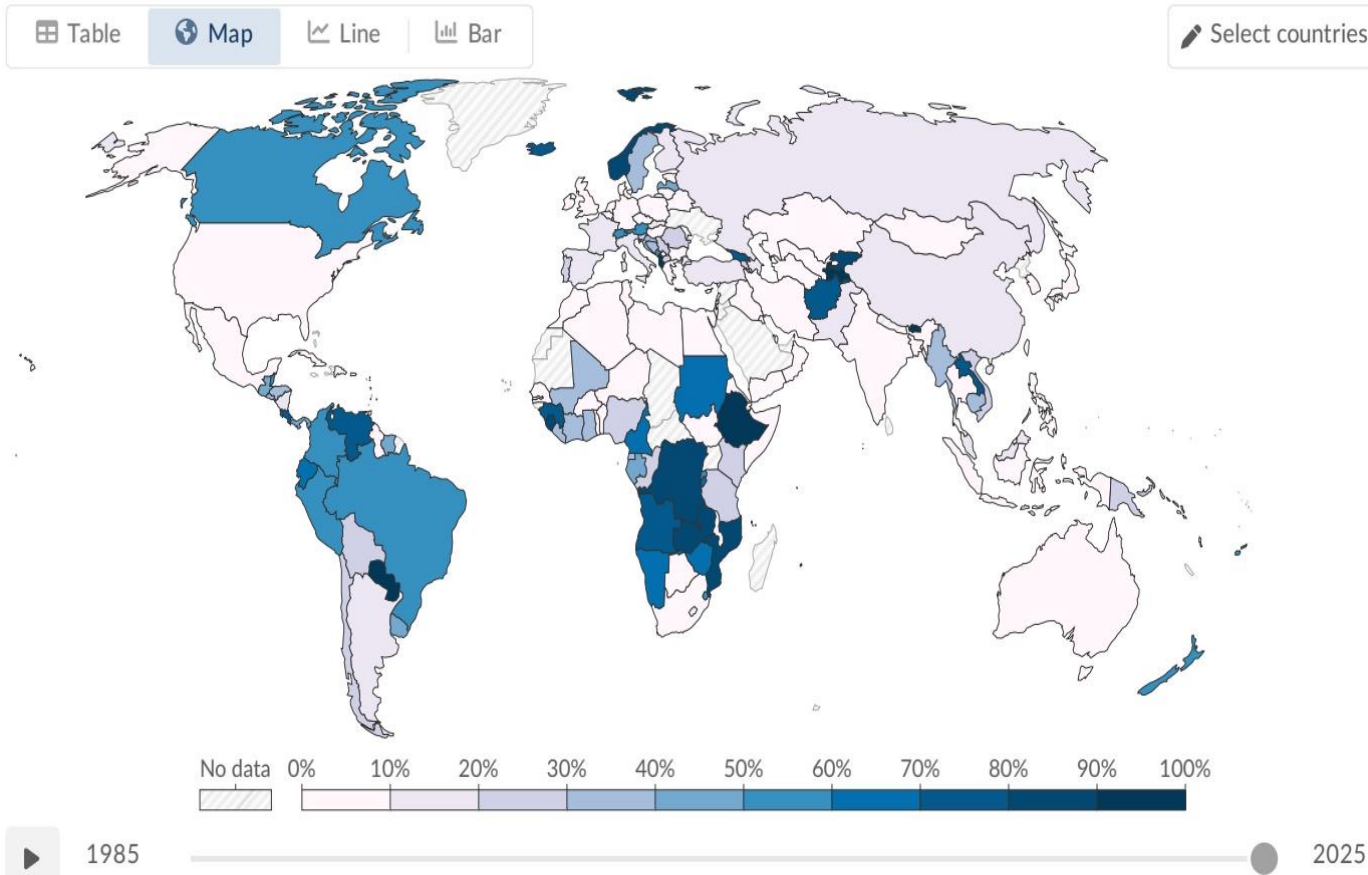
~21% of renewable electricity

Fastest-growing source — dramatic growth since 2010

Wind + Solar combined: ~4,400 TWh — nearly matching hydropower's ~4,500 TWh

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia & Latin America Lead

The energy transition is highly uneven across regions



LEADING REGIONS (65–98% renewable electricity)

- ▶ Scandinavia & Brazil: abundant hydropower
- ▶ Denmark & Germany: heavy wind and solar investment
- ▶ Canada: ~65% renewable electricity

LAGGING REGIONS (<10% renewable electricity)

- ▶ Sub-Saharan Africa: minimal renewable infrastructure
- ▶ Middle East: almost entirely fossil-fuel dependent
- ▶ Saudi Arabia: <2% renewable electricity

The transition to renewables is highly uneven — geography, policy, and investment all play decisive roles.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

Summary of key findings and the path forward

1

~14% of primary energy, ~30% of electricity

Renewables have reached a significant milestone, but the share in primary energy remains far lower due to hard-to-decarbonize transport and heating sectors.

2

Hydropower still dominates at ~50%

Hydro generates roughly half of all renewable electricity, but its growth potential is limited compared to wind and solar.

3

Wind + solar nearly match hydro

Wind (~2,400 TWh) and solar (~2,000 TWh) are the fastest-growing energy technologies in history and combined now nearly match hydropower's output.

4

Stark regional disparities persist

Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%.

5

The transition must accelerate

Current growth is encouraging but far from sufficient for climate targets. Accelerating deployment in lagging regions and hard-to-abate sectors is essential.

The world must accelerate the shift to renewables — especially in transport, heating, and regions currently left behind.